AI Companions and Mental Health: A Bibliometric Analysis

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1. Problem Background

AI-powered chatbots and virtual companions have gone from niche research projects to mainstream consumer products remarkably fast. Apps like Replika, Woebot, and a growing range of character-based chatbots now have tens of millions of users who interact with them for emotional support, conversation, and, increasingly, mental health help. This creates a pressing and underexplored question: what does regular use of these tools actually do to people's wellbeing?

The question matters on several levels. Scientifically, AI companionship sits at the crossroads of psychology, computer science, and public health, and the field still lacks basic consensus on how to measure effects or what outcomes even count. There is no agreed framework for when a chatbot's influence on a user becomes clinically relevant. For policymakers, the challenge is even more acute: regulators are being asked to govern tools they barely understand, with no established clinical standards for deployment in mental health contexts. At a societal level, millions of people, many of them young, isolated, or struggling with mental illness, are already using these systems as a substitute for human connection, with largely unknown consequences.

Several knowledge gaps make the problem hard to study. Most research to date captures only short-term snapshots; longitudinal studies that track real psychological outcomes over months are rare. Almost all empirical work has been done with healthy, educated, Western user samples rather than with clinical populations. Terminology is inconsistent across disciplines, making it hard to compare results. And despite the greatest unmet need for mental health support existing in low-income countries, almost no research comes from those settings.

A bibliometric analysis is well-placed to map the current state of this field, identify the key contributors and thematic clusters, and make the gaps visible.

2. Research Objectives

This bibliometric analysis has four goals: (1) to describe basic publication patterns, including how the number of articles has grown over time and how citations are distributed across years and sources; (2) to map which countries, research teams, and individual authors are most active; (3) to identify the most influential journals and the most-cited papers that define the field's intellectual foundation; and (4) to reveal current research hotspots and likely future directions by analyzing keyword patterns and thematic clusters in the literature.

3. Data Source and Scope

The data come from Dimensions (Digital Science), a large interdisciplinary academic database. It was chosen over alternatives like Web of Science or Scopus because it covers a broader range of publication types, including preprints and conference papers alongside journal articles. This matters for a fast-moving field where important findings often circulate before formal peer review. The dataset contains 1,462 articles published between 2010 and 2026, drawn from 886 different journals and outlets, with an average annual growth rate of 30.87%.

Only article-type documents were included. Records without author information were dropped during the analysis in R. The search combined two concept groups using Boolean logic: AI technology terms (AI chatbot, social chatbot, generative AI, conversational AI, virtual companion, artificial intimacy, etc.) AND psychological or social outcome terms (wellbeing, mental health, loneliness, social isolation, attachment, addiction, dependency, problematic use, etc.). Because Dimensions has no built-in language filter, a review of source titles and abstracts confirmed that English-language publications make up the large majority of the corpus.

4. Bibliometric Tools and Indicators

Three software environments were used. R with the bibliometrix package handled all descriptive statistics: annual publication counts, citation totals and averages, country and author productivity rankings, and the authors' production over time visualization. Core indicators include: total publications per year (TP), total citations (TC), average citations per year (TC/Y), annual growth rate (AGR), and the h-index – the largest number h such that an author has at least h papers each cited at least h times. These reflect both productivity and sustained impact.

VOSviewer was used for network construction and visualization. It produced three types of maps: co-authorship networks (connecting authors who have published together, weighted by how often), co-citation networks (connecting authors who are frequently cited together in the same papers, revealing shared intellectual territory without necessarily requiring direct collaboration), and bibliographic coupling maps (connecting papers or countries that share common references, meaning they draw on the same intellectual base). In all maps, node size reflects publication or citation volume, link thickness reflects tie strength, and color marks cluster membership determined by VOSviewer's community detection algorithm with resolution parameter 1.0.

ggplot2 in R was used for all bar charts and timeline plots, including annual production, citation averages, most productive authors and countries, and production-over-time charts.

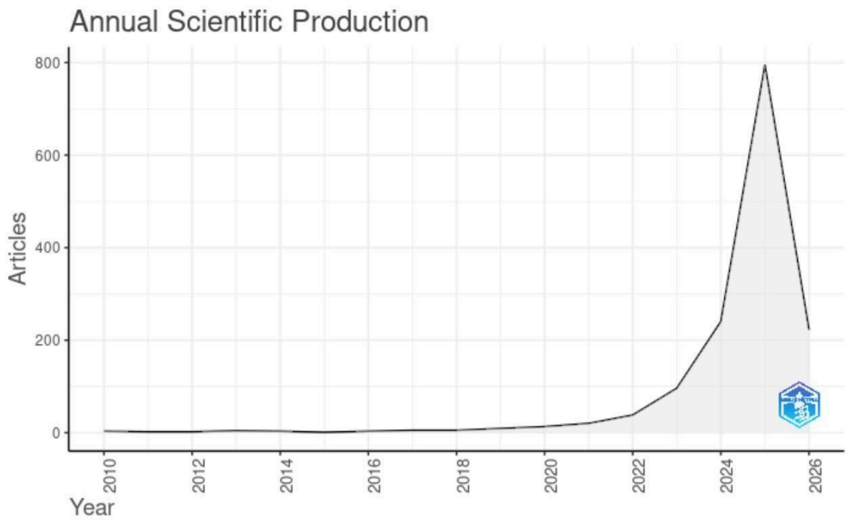
5. Analytical Framework

The analysis is organized in four steps. First, descriptive trends: annual publication volume and citation dynamics over time. Second, source, country, and author analyses: who produces this research and where. Third, co-authorship and co-citation network analysis: how researchers cluster into

teams and intellectual schools. Fourth, keyword and bibliographic coupling analysis: what themes dominate the field and how research fronts cluster.
For network filtering, a minimum of 5 documents per author was required before including that author in the co-authorship map. A minimum of 10 co-citations was applied for the co-citation network. For country-level bibliographic coupling, only countries with at least 5 publications were included. For keyword co-occurrence, only terms appearing at least 5 times were retained. A cluster is classified as a "hot topic" when it contains recent (post-2022) papers with high early citation rates and rapidly growing keyword frequency. A cluster is treated as an "emerging theme" when it consists primarily of recent papers that are accumulating attention but whose citation counts have not yet stabilized.

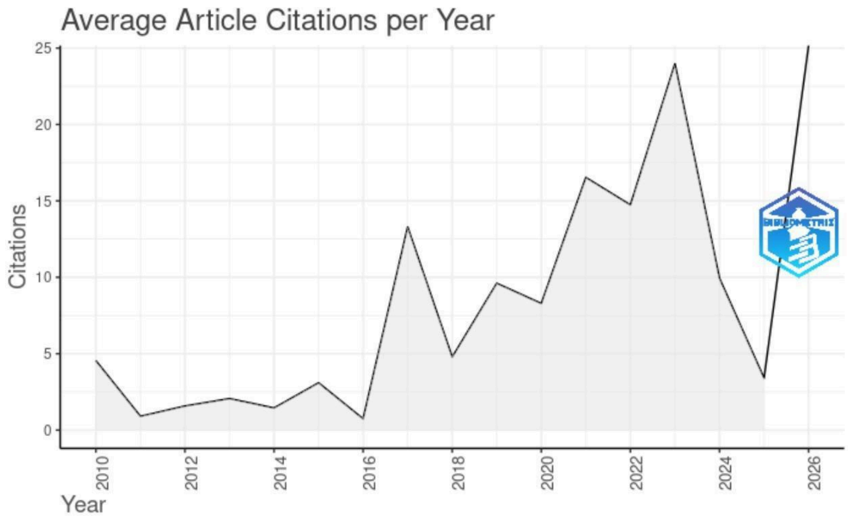
6. Publication and Citation Trends

The annual publication chart shows a field moving through three phases.



From 2010 to 2021, output was essentially flat at fewer than five articles per year. This reflects a pre-paradigm stage where a small number of researchers were developing foundational concepts without yet forming a recognizable research community. From 2019 to 2022, publications began rising steadily as commercial chatbots improved and gained users. After 2022, the curve shoots nearly vertical, peaking in 2025. This acceleration closely tracks the public release of ChatGPT in late 2022 and the wave of large language model-based companion apps that followed. A 30.87% compound annual growth rate is exceptionally high; for comparison, established fields like climate change or cancer research typically grow at 5-10% annually. This suggests the field is still in an explosive early expansion phase, not approaching maturity.

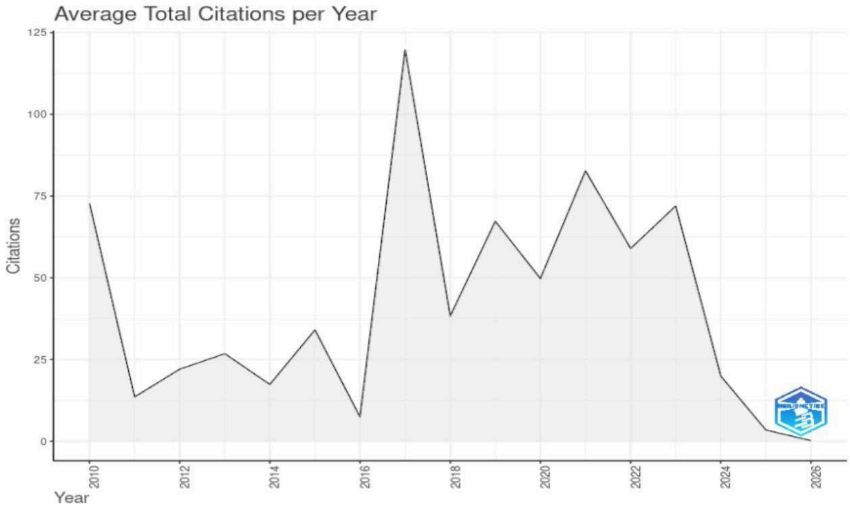
The average article citations per year graph indicates that post-2016 articles accumulate citations more rapidly, reflecting high immediate impact.



The average total citations per year graph shows the highest values for articles in

The average citations-per-year graph shows that papers published after 2016 accumulate citations faster than earlier work. In most mature fields, older papers with more time to accumulate references tend to have the highest citation counts. The reverse pattern here means that recent work is actively displacing older studies as the primary reference base. The field is not building steadily on its past; it is regularly rewriting its own foundations, which is a marker of a field still searching for paradigmatic stability.

2010 and for articles published between 2016 and 2024, identifying them as foundational works. Lower averages for recent years are expected due to less citation accumulation time.



The total citations chart shows two peaks: one around 2010 (a small number of early conceptual papers that defined key terms) and a broader peak covering 2016-2024 (an empirically richer wave of studies that now form the field's main reference base). The sharp drop for 2025 onward simply reflects citation immaturity – those papers have not had time to accumulate references. The two-wave pattern suggests the field has already gone through at least one significant paradigm shift, from early technical work on scripted chatbots toward research focused on large language models and their psychological effects.

7. Key Sources and Most Cited Works

The most cited papers in the corpus cluster around a few interconnected themes. The single most-cited paper is Ayers et al. (2023), which examines AI chatbots in medicine, followed by Park and Whang (2022) on empathy in human-robot interaction, and Park et al. (2022) on how chatbot identity disclosure affects willingness to donate. Below is the full top-20 ranking.

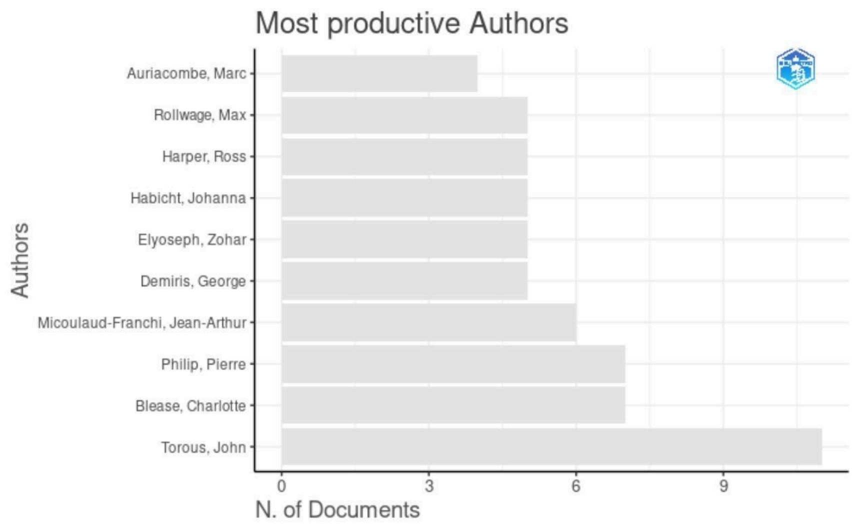
Top 20 Most Cited Papers

Authors	Year	Title	Citations	Source
Altamimi, Ibraheem; Altamimi, Abdullah; Alhumimidi, Abdullah S; Altamimi, Abdulaziz; Temsah, Mohamad-Hani	2023	Artificial Intelligence (AI) Chatbots in Medicine: A Supplement, Not a Substitute	98	Cureus
Park, Sung; Whang, Mincheol	2022	Empathy in Human–Robot Interaction: Designing for Social Robots	98	International Journal of Environmental Research and Public Health
Park, Gain; Yim, Myungok Chris; Chung, Jiyun; Lee, Seyoung	2022	Effect of AI chatbot empathy and identity disclosure on willingness to donate: the mediation of humanness and social presence	96	Behaviour and Information Technology
Cruz-Gonzalez, Pablo; He, Aaron Wan-Jia; Lam, Elly PoPo; Ng, Ingrid Man Ching; Li, Mandy Wingman; Hou, Rangchun; Chan, Jackie Ngai-Man; Sahni, Yuvraj; Guasch, Nestor Vinas; Miller, Tiev; Lau, Benson Wui-Man; Vidaña, Dalinda Isabel Sánchez	2025	Artificial intelligence in mental health care: a systematic review of diagnosis, monitoring, and intervention applications	94	Psychological Medicine
Ferraro, Carla; Demsar, Vlad; Sands, Sean; Restrepo, Mariluz; Campbell, Colin	2024	The paradoxes of generative AI-enabled customer service: A guide for managers	94	Business Horizons
Park, Gain; Chung, Jiyun; Lee, Seyoung	2022	Effect of AI chatbot emotional disclosure on user satisfaction and reuse intention for mental health counseling: a serial mediation model	93	Current Psychology
Tielman, Myrthe L.; Neerincx, Mark A.; Bidarra, Rafael; Kybartas, Ben; Brinkman, Willem-Paul	2017	A Therapy System for Post-Traumatic Stress Disorder Using a Virtual Agent and Virtual Storytelling to Reconstruct Traumatic Memories	92	Journal of Medical Systems
Gao, Yang; Dai, Yangbin; Zhang, Guangtao; Guo, Honglei; Mostajeran, Fariba; Zheng, Binge; Yu, Tao	2025	Trust in Virtual Agents: Exploring the Role of Stylization and Voice	9	IEEE Transactions on Visualization and Computer Graphics
Westerbeek, Hans	2025	Algorithmic fandom: how generative AI is reshaping sports marketing, fan engagement, and the integrity of sport	9	Frontiers in Sports and Active Living
Shankar, Ravi; Bundele, Anjali; Yap, Amanda; Mukhopadhyay, Amartya	2025	Development and feasibility testing of an AI-powered chatbot for early detection of caregiver burden: protocol for a mixed methods feasibility study	9	Frontiers in Psychiatry
Liu, Jiahui	2025	When Usefulness Fuels Fear: The Paradox of Generative AI Dependence and the Mitigating Role of AI Literacy	9	International Journal of Human-Computer Interaction
Weinstein, Netta; Itzchakov, Guy; Maniaci, Michael R.	2025	Exploring the connecting potential of AI: Integrating human interpersonal listening and parasocial support into human-computer interactions	9	Computers in Human Behavior Artificial Humans
Mahari, Robert; Pataranutaporn, Pat	2025	Addictive Intelligence: Understanding Psychological, Legal, and Technical Dimensions of AI Companionship	9	MIT Case Studies in Social and Ethical Responsibilities of Computing
Smit, Michelle; Wagner, Reinhard F.; Bond-Barnard, Taryn Jane	2025	Ambiguous regulations for dealing with AI in higher education can lead to moral hazards among students	9	Project Leadership and Society
Volpato, Riccardo; DeBruine, Lisa; Stumpf, Simone	2025	Trusting emotional support from generative artificial intelligence: a conceptual review	9	Computers in Human Behavior Artificial Humans
Shrestha, Buddhi Laxmi Lakhe; Dahal, Niroj; Hasan, Kamrul; Paudel, Santosh; Kapar, Hiralal	2025	Generative AI on professional development: a narrative inquiry using TPACK framework	9	Frontiers in Education
Hu, Fuwen; Wang, Chun; Wu, Xuefei	2025	Generative Artificial Intelligence-Enabled Facility Layout Design Paradigm	9	Applied Sciences
Madsen, Dag Øivind; Toston, David Matthew	2025	ChatGPT and Digital Transformation: A Narrative Review of Its Role in Health, Education, and the Economy	9	Digital
Alalwan, Ali Abdallah; Algharabat, Raed; Abu El Samen, Amjad; Albanna, Hanaa; Al-Okaily, Manaf	2025	Examining the impact of anthropomorphism and AI-chatbots service quality on online customer flow experience – exploring the moderating role of telepresence	9	Journal of Consumer Marketing
Sheehy, Lisa; Bouchard, Stéphane; Kakkar, Anupriya; Hakim, Rama El; Lhoest, Justine; Frank, Andrew	2024	Development and Initial Testing of an Artificial Intelligence-Based Virtual Reality Companion for People Living with Dementia in Long-Term Care	9	Journal of Clinical Medicine

Several patterns stand out. First, the most-cited works are dominated by papers published between 2022 and 2025, confirming that the field's intellectual core has been recently established rather than inherited from older work. Second, three distinct themes appear in the top 20: chatbot empathy and humanness (how anthropomorphic design affects user trust and behavior), clinical applications (systematic reviews on AI in mental health, dementia support, caregiver tools), and the risks and paradoxes of AI companions (dependency, addiction, trust issues, regulatory ambiguity). Third, the top-cited author by publication count, Torous (John), does not appear in the highest-citation papers – the most-cited works are often single influential papers by authors who are not among the most prolific. This disconnect between productivity and impact is common in fast-moving fields where one well-timed paper can define an agenda.

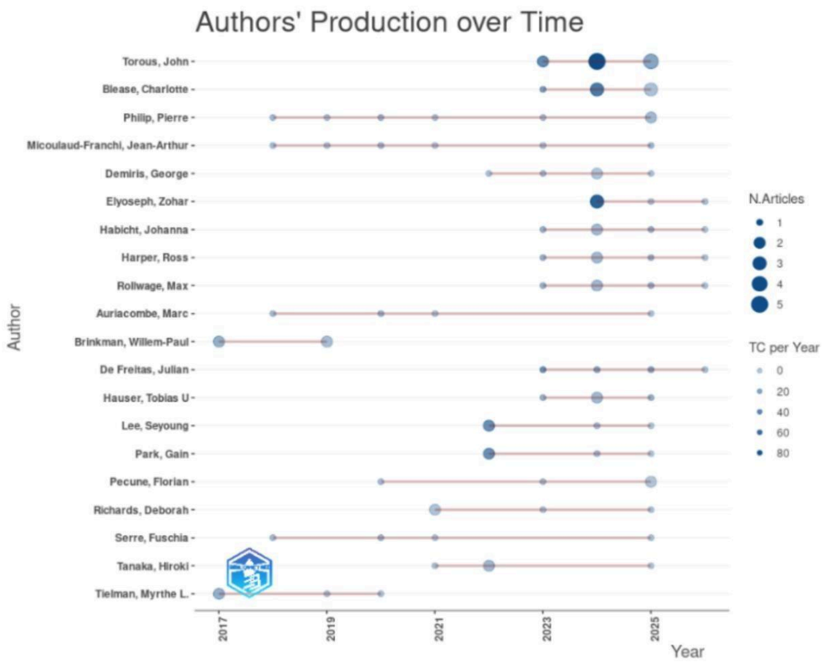
8. Authors and Research Teams

The corpus involves 4,978 authors across 1,462 articles. The chart below shows the 10 most productive by publication count.



The authors' production over time plot shows that most authors began publishing around 2020–2022, with a clear increase in activity through

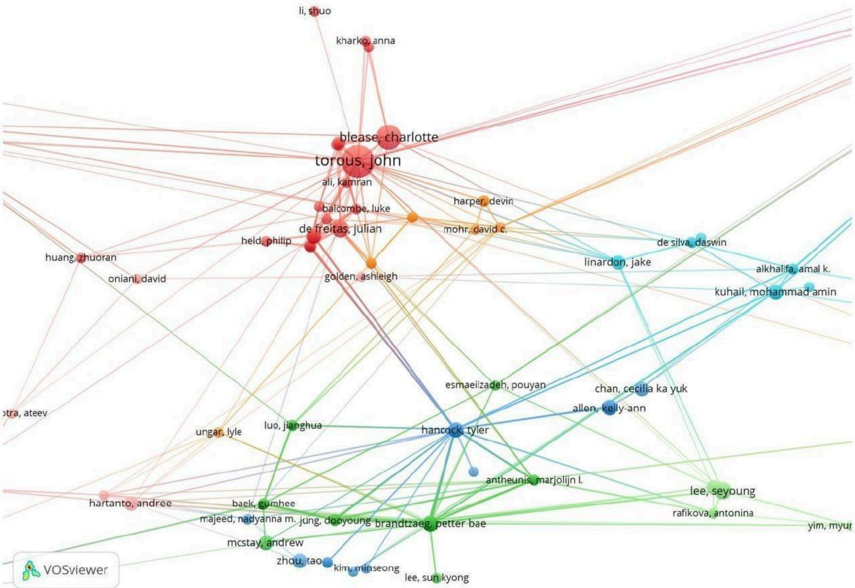
Torous, John leads by a clear margin with approximately 9 publications, putting him well ahead of the next group. His work centers on digital psychiatry and mobile mental health technology. Blease, Charlotte and Philip, Pierre both have around 6-7 publications. The remaining top-10 authors: Elyoseph, Zohar; Habicht, Johanna; Harper, Ross; Rollwage, Max; Demiris, George; Micoulaud-Franchi, Jean-Arthur; and Auriacombe, Marc - reflect the field's dual disciplinary base in clinical psychology and human-computer interaction.



Most cited papers plot, however, show a different ranking. The most impactful

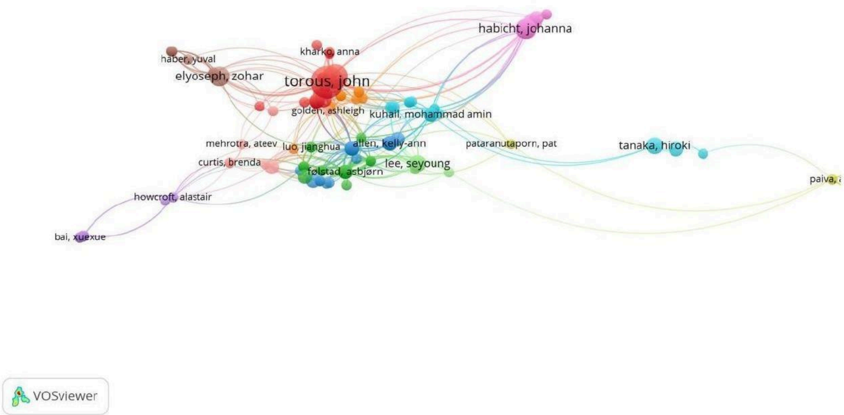
The production-over-time chart shows that most of these researchers only started publishing in this area around 2020-2022, consistent with the broader acceleration of the field. The dot sizes encode annual citation counts: Torous and Elyoseph stand out with the largest dots, meaning their papers attract disproportionately high citations relative to their volume. Early contributors like Brinkmann and Tielman, who were active before 2019, appear to have become less central – their work focused on early scripted chatbot prototypes that have since been superseded. The field appears to have shifted from engineering-oriented early work toward applied clinical and behavioral research requiring different expertise.

9. Co-authorship and Co-citation Networks



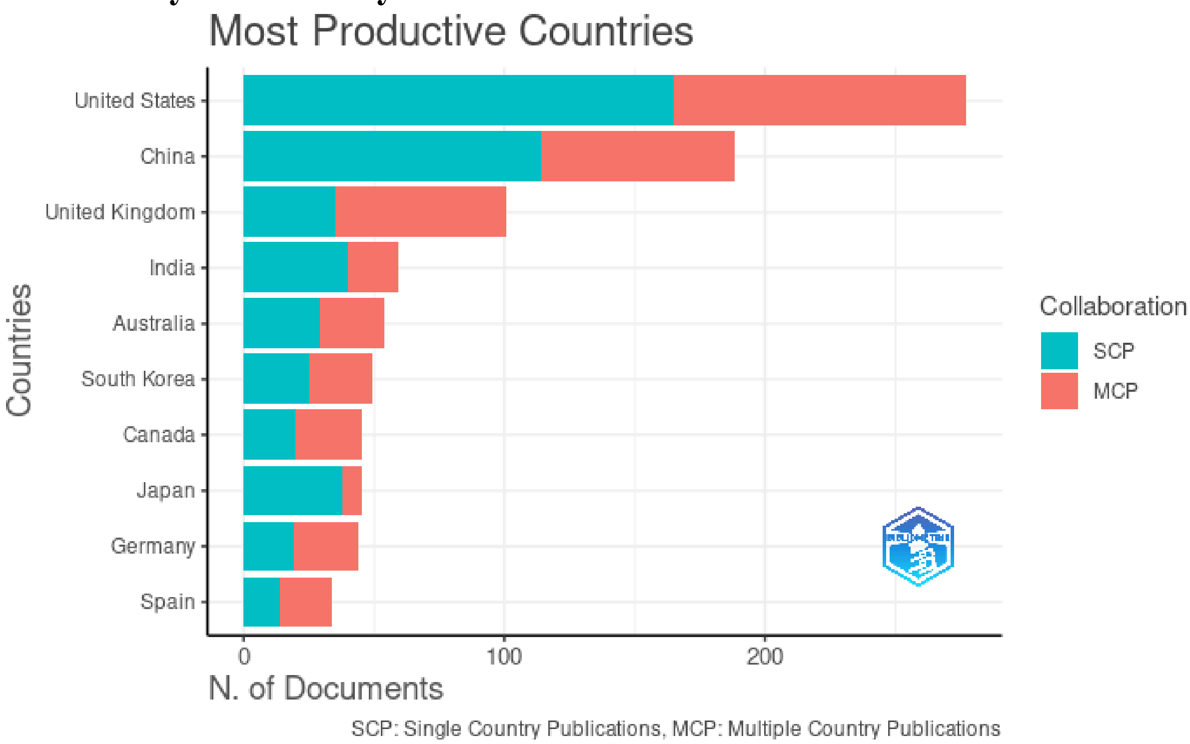
The dataset comprises 4,978 authors contributing to the 1,462 articles analyzed. Most productive authors (based on publication count) are led by Thorous John. He is the author of significantly more papers than all other

The co-authorship network shows three distinct clusters with limited connections between them. The largest, centered on Philip, Pierre and De Sevin, Etienne, includes Auriacombe, Marc; Serre, Fuschia; Dupuy, Lucile; Baillet, Emmanuelle; and Bonhomme, Emilien – a tightly connected French research group focused on psychiatric applications of AI tools, particularly in addiction medicine and mental health services. A second smaller cluster around Pecune, Florian and Coelho, Julien works more on human-robot interaction and social agent design. A third cluster, including Amorim, Michel-Ange; Brunet-Gouet, Eric; and Bazin, Nadine, represents a separate French psychiatry team. The three clusters connect only weakly, showing that even researchers working on closely related problems have not formed cross-group collaborations. This is a sign of a field that is still fragmented into parallel research silos rather than an integrated community.
The co-citation network below reveals a very different picture of the field's intellectual structure.

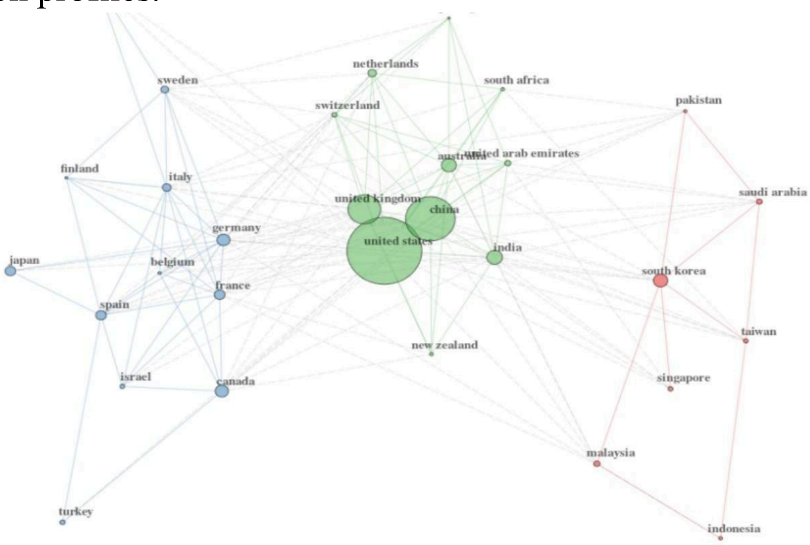


In the co-citation map, Torous, John is the most central node – his papers are cited alongside more others than anyone else in the corpus, making him the field's primary intellectual anchor. He is flanked by Elyoseph, Zohar; Habicht, Johanna; Fjelstad, Asbjorn; and others who form a dense central cluster. These authors represent the shared foundation that most new papers draw on. Pataranutaporn, Pat; Tanaka, Hiroki; and Paiva, A. sit at the periphery, representing specialized sub-areas cited selectively rather than as general references. A structurally important finding is the mismatch between the two maps: the French-language cluster that dominates the co-authorship network is nearly absent from the co-citation core. This means the French group, while internally very collaborative, has not yet become a primary reference point for the wider international field. Their work is being produced but not yet widely integrated into the mainstream intellectual conversation.

10. Country Productivity and Collaboration



The United States leads all countries in total publications by a substantial margin, followed by China and the United Kingdom. The bar chart distinguishes Single Country Publications (SCP, teal) from Multiple Country Publications (MCP, red). The US and China both have high SCP proportions, meaning much of their output is produced domestically. The UK shows a higher MCP share, reflecting a stronger orientation toward international partnerships. India and Japan are productive but primarily domestic. Australia, South Korea, Canada, Germany, and Spain round out the top 10, each with moderate output and mixed collaboration profiles.



The country collaboration network reveals three clusters. The first is anchored by the United States, which has more international connections than any other country, acting as the main broker between different regional research communities. The US-China-UK triangle forms the core, with Canada, Australia, the Netherlands, Switzerland, Portugal, South Africa, India, and New Zealand also integrated into this cluster. These countries are part of the global research mainstream. The second cluster is a group of European countries, mostly continental, with dense ties to each other – a pattern likely facilitated by shared EU funding programs and geographic proximity. The third cluster groups South Korea, Singapore, Saudi Arabia, Indonesia, Malaysia, Taiwan, and Pakistan, suggesting a distinct regional network in Asia and the Middle East, possibly driven by shared national AI research initiatives. One critical observation: Africa (except South Africa), Latin America, and Eastern Europe are almost entirely absent from the network. These are significant gaps given that the populations most likely to benefit from AI mental health tools – those with limited access to human clinicians – are concentrated in exactly these regions.

11. Terms, Keywords, Hotspots, and Themes

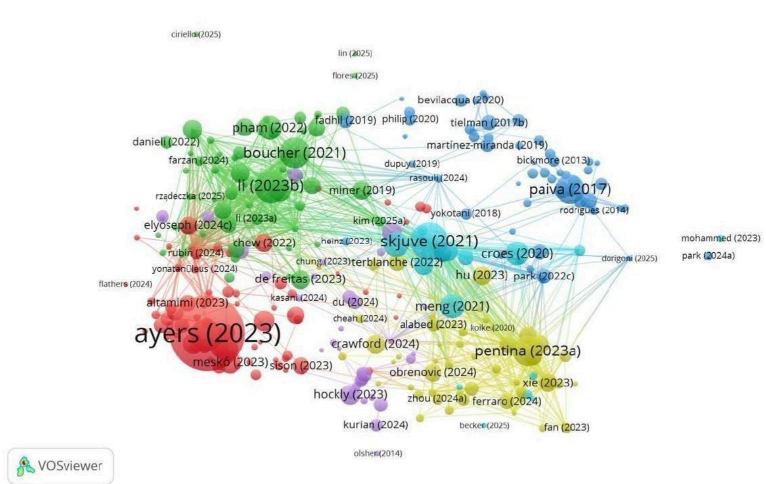
The table below lists the top terms by frequency extracted from titles and abstracts across the full corpus.

Keyword	Frequency	Percentage
communication	123	2.49
technological	81	1.64
personalization	74	1.50
comprehensive	69	1.40
professionals	62	1.26
transformative	62	1.26
interactions	61	1.24
characteristics	59	1.20
accessibility	53	1.07
crosssectional	52	1.05
companionship	48	0.97
collaboration	43	0.87
transformation	42	0.85
anthropomorphism	41	0.83
interdisciplinary	40	0.81
technologies	40	0.81
institutional	37	0.75
classification	35	0.71
identification	34	0.69
anthropomorphic	33	0.67

The most frequent term is "communication" (123 occurrences, 2.49%), followed by "technological" (81, 1.64%), "personalization" (74, 1.50%), "comprehensive"

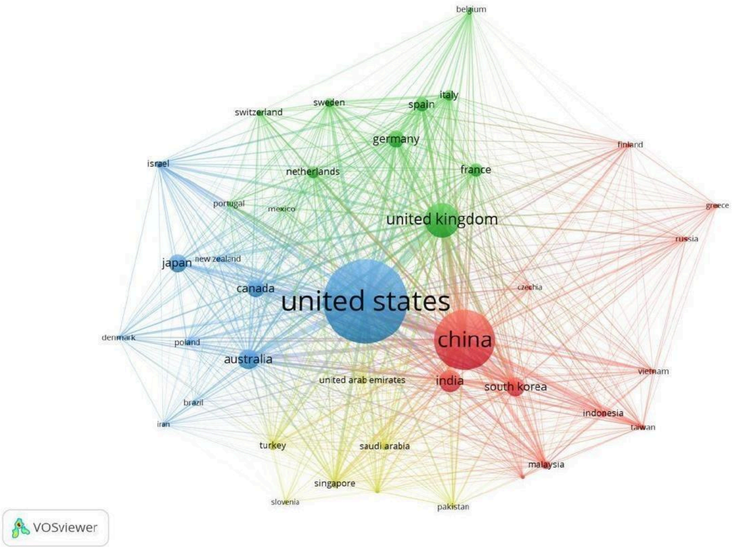
(69, 1.40%), and "professionals" and "transformative" (62 each, 1.26%). Further down: interactions, characteristics, accessibility, cross-sectional, companionship, collaboration, anthropomorphism, and interdisciplinary. A notable problem is that several high-frequency terms – particularly "comprehensive," "characteristics," "identification," and "institutional" – are generic academic filler words rather than field-specific concepts. Their high frequency reflects the way abstracts are written rather than meaningful thematic content. This points to a terminological immaturity in the field: researchers have not yet converged on a stable shared vocabulary. In contrast, genuinely substantive terms like "personalization," "companionship," "anthropomorphism," and "accessibility" do reflect real thematic content and align with the clusters visible in the network maps.
The bibliographic coupling map groups papers by shared reference pools, revealing four distinct thematic clusters.

bibliographic coupling



Cluster 1 (red, largest): centered on Ayers (2023), Mesko (2023), Hockly (2023), Crawford (2024), and Ferraro (2024). This cluster covers the deployment of generative AI tools in clinical and professional settings – healthcare workers, educators, managers. It is the currently dominant and fastest-growing research area, driven by the rapid adoption of ChatGPT-class tools across professions. Cluster 2 (green): anchored by Skjuve (2021), Croes (2020), and Paiva (2017). These papers study how users form relationships with chatbots – attachment formation, the role of chatbot personality and empathy, loneliness reduction – representing the social psychology strand of the field. This cluster also includes early work on virtual companions in therapeutic contexts. Cluster 3 (blue/orange, right side): around Pentina (2023a), Xie (2023), Mohammed (2023), and Park (2024a). This cluster examines AI companions in consumer and marketing contexts – brand chatbots, customer service, and the commercial dimensions of AI relationships. Cluster 4 (smaller, upper area): including Tielman (2017b), Martinez-Miranda (2018), Rodrigues (2018), and Bickmore (2013). This represents the foundational technical

literature on therapeutic virtual agents and health counseling systems – the original wave of research that preceded the generative AI era and now serves as a historical reference base.
The bibliographic coupling by country confirms the geographic structure identified earlier.



The US, China, and UK dominate as the largest nodes, but China appears nearly as large as the US in terms of shared reference pools, suggesting Chinese research is deeply embedded in the field's intellectual infrastructure even though individual Chinese authors are less visible in the co-citation network. The map also shows that the European cluster and the Asian cluster each share internal reference pools that partially overlap with but remain distinct from the US-UK-China core, pointing to some degree of regional intellectual differentiation in research approaches.

12. Overall State of the Field

The four thematic clusters each address a different dimension of the same underlying problem. Clinical application research tests whether AI companions can deliver real mental health benefits at scale. Social and relational research examines the psychological mechanisms by which users form bonds with AI systems, and what that means for loneliness, attachment, and social behavior. Risk and ethics research provides the normative framework for governing these tools responsibly. Design and technical research translates insights from the other clusters into functional systems.
In principle, these clusters cover the problem well. In practice, the field has three structural imbalances that limit its usefulness. The first is geographic.

Research is concentrated in a small set of high-income countries. The populations most likely to benefit from AI mental health tools – people in low-income countries with few human therapists – are nearly absent from the research. This means current findings may not generalize to the contexts where these tools are most urgently needed, and current design choices reflect the preferences of Western, educated user samples.

The second imbalance is methodological. The corpus is dominated by cross-sectional surveys and short-term user experience studies. These can show correlations and immediate reactions, but they cannot answer the most important question: does sustained use of an AI companion help or harm people over time? Randomized controlled trials are essentially absent. This is a critical gap because the policy and regulatory decisions being made right now about AI mental health tools are being made without the kind of evidence that clinical practice normally requires.

The third imbalance is disciplinary. The French-language psychiatric research cluster and the English-language clinical/HCI mainstream are not citing each other despite working on closely related topics. Similarly, the consumer behavior cluster and the clinical cluster rarely reference each other, even though understanding how users psychologically engage with AI companions is relevant to both. This fragmentation slows knowledge accumulation and increases the risk of parallel duplicated effort.

13. Methodological Gaps and Limitations

Dimensions has genuine advantages for this analysis, particularly its broad coverage of preprints and conference proceedings that traditional databases miss. However, it has limitations. Its lack of a built-in language filter means the English-language bias of the corpus cannot be fully controlled. The Boolean search query is necessarily imprecise – it will include some papers that mention these terms only incidentally and miss others that study the same phenomena using different terminology. Future searches could improve precision by incorporating controlled vocabulary terms once the field has developed more stable indexing standards.

VOSviewer's clustering depends on a resolution parameter set by the researcher. Different settings produce meaningfully different cluster structures, so the four thematic clusters identified here represent one defensible interpretation of the data, not a ground truth. Publication counts and citation frequencies measure scholarly attention, not research quality or real-world impact.

Several methodological improvements could strengthen future work in this area. First, using topic modeling on full abstracts (LDA or BERTopic) rather than author keywords alone would provide richer and less noisy thematic clustering. Second, combining data from multiple databases would reduce single-source coverage bias. Third, including grey literature – clinical guidelines, policy documents, regulatory filings – would better capture the practical dimensions of the field that journal articles often miss. Fourth, tracking citation network evolution over multiple time points rather than a single snapshot would allow the detection of paradigm shifts in real time.

14. Summary

This analysis mapped 1,462 articles on AI companions and mental health published between 2010 and 2026, retrieved from the Dimensions database using a Boolean search combining AI technology and psychological outcome terms. The field has grown at 30.87% per year on average, with a dramatic post-2022 acceleration driven by the arrival of large language model-based chatbots. Publication and citation dynamics reveal two waves of foundational influence: early conceptual work from around 2010, and a second empirical wave from 2016-2024 that now constitutes the primary reference base.

The most productive author is Torous, John, who also sits at the center of the co-citation network. Co-authorship analysis reveals a fragmented structure built around three clusters, with the largest being a French psychiatric research group that is internally cohesive but poorly integrated into the wider international field. The US, China, and UK dominate both output and the shared intellectual infrastructure, while the Global South is almost entirely absent.

Bibliographic coupling and keyword analysis identified four core thematic areas: clinical and professional AI tool deployment (currently the fastest-growing cluster), social and relational dynamics of human-AI interaction (the social psychology strand), risks and ethics including dependency and regulation, and foundational technical work on therapeutic virtual agents. The most rapidly growing topics involve large language model deployment in mental health contexts, emotional dependency, and regulatory uncertainty.

The field faces three structural problems that limit its scientific and practical value: heavy reliance on short-term, cross-sectional methodology; geographic concentration that excludes the populations most in need; and disciplinary fragmentation that prevents knowledge from flowing between research communities. Addressing these gaps – through longitudinal clinical trials, globally inclusive research designs, and deliberate cross-disciplinary integration – is the most important next step for this field.